

Harshiddhi Pathak

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Summary

PhD researcher in the Artificial Intelligence discipline, Computer Science and Engineering Department, IIT Gandhinagar, supervised by Dr. Yogesh Kumar Meena. My work focuses on Brain-Computer Interfaces, using EEG and multimodal neuroimaging (MRI/fMRI) with deep learning to detect and classify neurodegenerative disorders such as Alzheimer's disease and dementia. I build interpretable, attention-based models that turn multimodal neural data into clinically meaningful insight, drawing on a biomedical engineering background and prior research in MRI-fMRI fusion.

Research Focus

- **Brain-Computer Interfaces:** EEG signal acquisition and decoding for neurodegenerative disorder classification and early diagnosis
- **Multimodal Medical Imaging:** Cross-attention based fusion of MRI and fMRI for early Alzheimer's disease detection
- **Deep Learning for Neuroscience:** Graph Attention Networks with Synchrosqueezed Chirplet Transform (SCT)-derived EEG features for Alzheimer's and dementia classification
- **Biomedical Signal Processing:** Sensor-based physiological monitoring devices and biomedical instrumentation for clinical and diagnostic applications

Education

Doctor of Philosophy (PhD), Artificial Intelligence Jul 2026 – Present

Indian Institute of Technology Gandhinagar
Supervisor: Dr. Yogesh Kumar Meena

Master of Technology, Biomedical Engineering Jul 2024 – Apr 2026

Indian Institute of Technology Kharagpur – CGPA: 9.09 / 10
Thesis: Cross-Attention Based Fusion of MRI and fMRI for Early Alzheimer's Disease Detection
Thesis Supervisors: Prof. Manjunatha Mahadevappa, Dr. Mrinal Acharya

Bachelor of Engineering, Biomedical Engineering Jul 2020 – Apr 2024

Gujarat Technological University (GTU), Ahmedabad – CGPA: 8.91 / 10
Focus: Biomedical Signals, Medical Imaging, Biomedical Instrumentation

Coursework

- **PhD Coursework:** Deep Learning, Human Computer Interaction, Natural Language Processing
- **M.Tech Coursework:** Digital Image Processing, Geometric Methods for Computer Vision, Signals and Systems, Biomedical Informatics, Mathematical Methods in Biomedical Engineering, MEMS and Biosensors.

Projects

Cross-Attention Based Fusion of MRI and fMRI for Early Alzheimer's Disease Detection

Designed and implemented a cross-attention based MRI-fMRI fusion model in PyTorch for Alzheimer's disease classification, with an end-to-end FSL preprocessing pipeline (BET, registration, VBM) and extraction of ROI-level gray matter and functional connectivity features. Achieved MRI accuracy of 91%, fMRI accuracy of 88.95%, and ROC-AUC of 0.90 using self-attention and cross-attention architectures.

Graph Attention Networks with SCT-Derived EEG Features for Alzheimer's and Dementia Classification

Built a GAT-based deep learning framework using Synchrosqueezed Chirplet Transform (SCT) EEG features

to model spatio-temporal dependencies for neurodegenerative disorder classification. Achieved 92.31% accuracy (AD vs. CN) and 83.33% multi-class accuracy (AD/CN/FTD).

Bone Segmentation from X-ray Images

Benchmarked Watershed, Superpixel, and U-Net architectures on the MURA dataset for bone segmentation, achieving a Dice score of 0.978 using U-Net.

EMG Measurement Device using Pressure Sensors

Designed and validated a flex-sensor-based EMG measurement device for diabetic neuropathy monitoring.

Experience

Teaching Assistant, Machine Learning – IIT Gandhinagar

Aug'26 – Present

Teaching Assistant, Biomedical Instrumentation Lab – IIT Kharagpur

Jul'25 – Nov'25

Assisted in course delivery, lab supervision, assignment design, and evaluation; supported students in understanding core concepts and practical implementation.

Research Intern, Neuromorphic Research Team, JSA LAABS Pvt. Ltd. (Remote) Jun'24 – Dec'25

Gained hands-on experience with Spiking Neural Networks (SNNs) and advanced attention mechanisms, working within complex codebase architectures in the domain.

Publications

1. Harshiddhi Pathak et al. “An Attention-Based Framework for Alzheimer’s Disease Classification Using Resting-State fMRI.” *IEEE EMBC 2026*, Toronto, Canada, Jul 26–30, 2026. (Presented)
2. “Graph Attention Networks with SCT-Derived EEG Features for Alzheimer’s and Dementia Classification.” Manuscript in preparation, targeting *Journal of Neural Engineering*.
3. “Cross-Attention Based Fusion of MRI and fMRI for Early Alzheimer’s Disease Detection.” (In preparation)

Technical Skills

- **Machine Learning:** PyTorch, Deep Learning, Attention Mechanism
- **Neuroimaging & Signal Processing:** EEG Processing, MRI/fMRI Processing, FSL Software Tool
- **Programming:** Python, c++, MATLAB
- **Domains:** Biomedical Instrumentation, Medical Imaging, Brain-Computer Interfaces

Certifications

- Machine Learning for Engineering and Science Applications – NPTEL (Jan–Apr 2025)
- Machine Learning – IIT Roorkee (Online Certification)

Conferences and Workshops Attended

IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI '25) 2025

Atlanta, Georgia – Data Competition: Predicting Macronutrient Information

Team “EDAers” – Global Finalist (Top 4). Project: Predicting Meal Carbohydrate Caloric Ratio (CCR) from 3-hour PPGR.

RadMed Conference 2025

Dec 17–19, 2025

School of Medical Science & Technology, IIT Kharagpur

Presented a poster titled: “EEG-Based Seizure Detection Using Chirplet Transform and Graph Attention Neural Networks.”